# Week 10 Milestone Report

Recommendation, Ranking, and Predictive Decision Engine

MovieLens Discovery System — Semester Project

Grupo 06 — Big Data

`BD-Grupo-06/movielens-discovery`

June 12, 2026

## Contents

# 1 Executive Summary

This report covers Week 10 of the semester project: building a recommendation and ranking system for the MovieLens 25M catalog. Three systems were built and evaluated offline using a genre-relevance protocol over 4,994 query movies:

Table 1: System overview and primary evaluation result (NDCG@10)

| **System** | **Type** | **Signal** | **NDCG@10** |
|---|---|---|---|
| `popularity_global` | Baseline | Bayesian rating count | 0.8390 |
| `content_cosine` | Baseline | Cosine sim. on AE-13 embeddings | **0.9925** |
| `svd_collaborative` | Advanced | Item factors from SVD ($k = 50$) | 0.9496 |

The content-based cosine model achieves the highest NDCG@10, driven by the dense, genre-coherent structure of the Week 7 autoencoder embedding space. The SVD collaborative model ranks second, demonstrating that behavioral co-occurrence provides complementary signal to content features.

# 2 Objective

The objective of Week 10 was to connect the prior technical layers to a concrete product decision:

*Given a seed movie, rank the catalog by relevance to that movie.*

This is a **ranking and discovery system** (item-to-item), appropriate for catalog discovery without a logged-in user context. Specific questions addressed:

- Does SVD collaborative filtering beat content-based similarity on genre-relevant recommendations?
- Where are the failure cases of each system?
- How does the Week 7 cluster layer support the recommendation layer?

# 3 Inputs and Reproducible Artifacts

## 3.1 Input artifacts from previous weeks

Table 2: Inputs consumed from prior weeks

| **Artifact** | **Source** | **Purpose** |
|---|---|---|
| `week07_autoencoder_embeddings_latent_13.parquet` | Week 7 | Content embedding (13-dim, 5 |
| `week07_kmeans_assignments.csv` | Week 7 | Cluster-aware popularity ($k =$ |
| `ratings_clean.parquet` | Week 3 | SVD and popularity (25M rat |
| `movies_catalog.parquet` | Week 3 | Metadata and genre labels (59 |

### 3.2 Week 10 output artifacts

Table 3: Output artifacts produced in Week 10 (31 files, ≈40 MB total)

| File | Description |
|---|---|
| `week10_popularity_global.csv` | Global Bayesian ranking (59,047 movies) |
| `week10_popularity_cluster.csv` | Cluster-aware ranking |
| `week10_content_recs_top20.parquet` | Content-based top-20 per query |
| `week10_svd_item_factors.parquet` | SVD item factors (13,176 × 50) |
| `week10_svd_recs_top20.parquet` | SVD collaborative top-20 per query |
| `week10_evaluation_results.csv` | Full metric comparison table |
| `week10_error_analysis.csv` | Strong and failure cases per system |
| `week10_evaluation_summary.json` | JSON summary of all evaluation results (includes LO |
| `week10_loo_evaluation_results.csv` | Leave-One-Out metric comparison table |
| `week10_loo_evaluation_comparison.png` | LOO bar chart (Hit Rate, Precision, NDCG@K) |

## 4 Baseline Systems

### 4.1 Global popularity baseline

Every movie is ranked by its Bayesian-adjusted score:

$$\text{bayesian\_score} = \frac{C \cdot \mu + n \cdot \bar{r}}{C + n} \tag{1}$$

**Parameters**: global mean $\mu = 3.0714$, prior count $C = 1.0$ (10th percentile of rating counts).

The catalog has a strong long-tail: 59,047 movies, but fewer than 5% have more than 10,000 ratings, as shown in Figure 1.

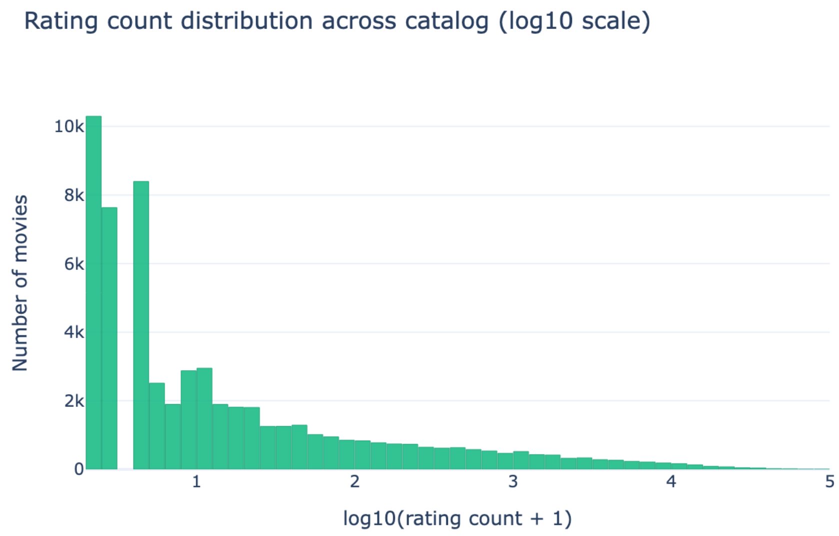


Figure 1: Rating count distribution across the catalog (log10 scale). The long-tail structure motivates the Bayesian adjustment: most movies have fewer than 100 ratings.

## 4.2 Cluster-aware popularity baseline

Restricts the ranking to movies within the same Week 7 cluster as the query movie. Table 4 shows the cluster engagement disparity: Cluster 4 (23,093 movies, mean 9.1 ratings each) represents the low-engagement catalog tail, while Cluster 0 averages 1,666 ratings per movie.

Table 4: Cluster-level popularity statistics

| Cluster | Movies | Mean rating count | Mean avg. rating | Mean Bayesian score |
|---:|---:|---:|---:|---:|
| 0 | 9,363 | 1,665.9 | 3.291 | 3.288 |
| 1 | 2,650 | 505.1 | 3.130 | 3.152 |
| 2 | 5,013 | 327.6 | 3.064 | 3.091 |
| 3 | 10,036 | 463.6 | 2.739 | 2.815 |
| 4 | 23,093 | 9.1 | 3.042 | 3.061 |
| 5 | 5,346 | 55.4 | 3.383 | 3.321 |
| 6 | 3,546 | 356.1 | 3.121 | 3.126 |

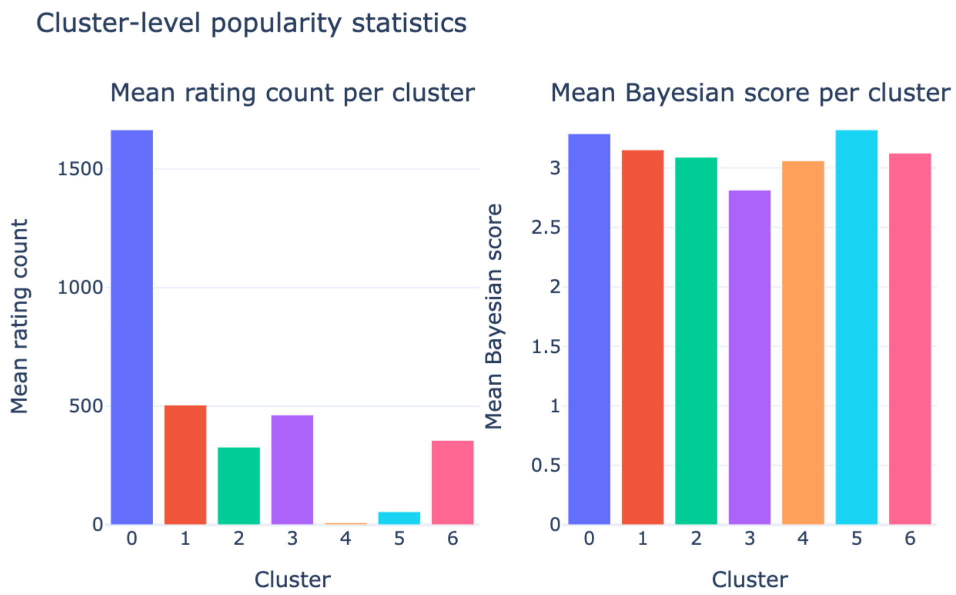


Figure 2: Mean rating count and Bayesian score per Week 7 cluster. Cluster 0 (mainstream Drama/Comedy) dominates engagement; Cluster 4 (long-tail catalog) is minimal.

# 5 Content-Based Baseline

## 5.1 Method

Top-20 most similar movies are retrieved using cosine similarity over the 13-dimensional autoencoder embedding space from Week 7:

$$\mathrm{sim}(i, j) = \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \cdot \|\mathbf{e}_j\|} \tag{2}$$

Embedding rows are L2-normalized before retrieval (dot product = cosine similarity).

## 5.2 Similarity distribution

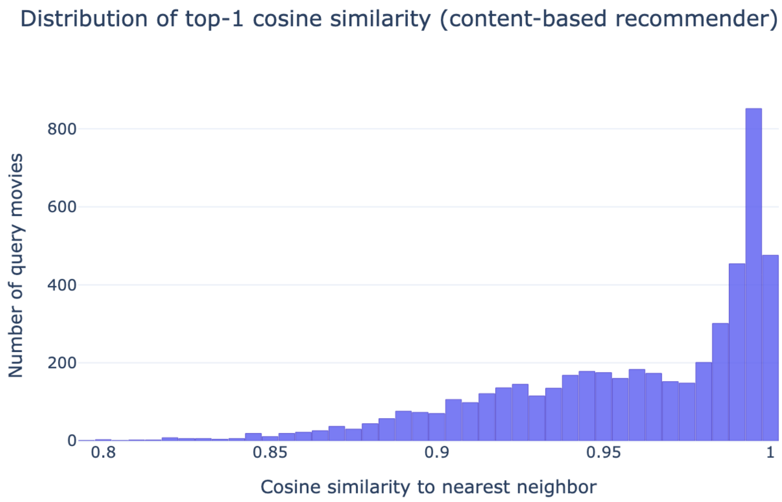


Figure 3: Distribution of top-1 cosine similarity (nearest neighbor) across 5,000 query movies. The concentration near 1.0 confirms the autoencoder learned a dense, genre-coherent embedding space.

# 6 Advanced System: SVD Collaborative Filtering

## 6.1 Model design

The advanced system factorizes the mean-centered user-item rating matrix:

$$\mathbf{R}_{\text{centered}} \approx \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^{\top} \tag{3}$$

**Matrix**: 162,540 users $\times$ 13,176 movies (after filtering: $\geq$ 50 ratings/movie, $\geq$ 20 ratings/user). **Density**: 1.15%. **Centering**: per-user mean subtraction. **Latent factors**: $k = 50$, chosen from the sweep below.

Recommendations use cosine similarity between L2-normalized rows of $\mathbf{V}$ (item factor vectors), capturing behavioral co-occurrence patterns.

## 6.2 Latent factor sweep

Table 5: SVD latent factor sweep results

| Components | Explained variance | Time (s) |
|---:|---:|---|
| 10 | 9.33% | 1.35 |
| 20 | 12.27% | 2.11 |
| 30 | 14.42% | 2.07 |
| **50** | **17.79%** | **2.83** |
| 75 | 21.11% | 3.57 |
| 100 | 23.89% | 4.60 |
| 150 | 28.58% | 5.37 |

$k = 50$ was selected: the variance curve flattens past this point with diminishing returns (Figure 4).

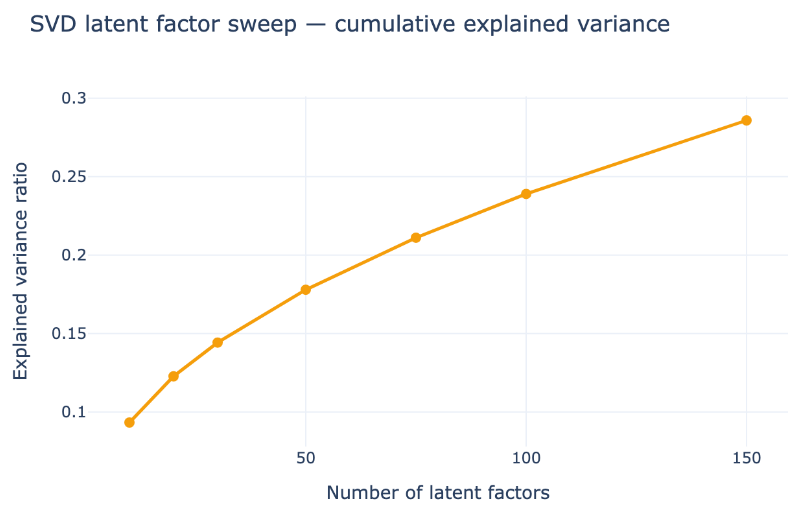


Figure 4: SVD cumulative explained variance vs. number of components. The elbow appears near $k = 50$.

## 6.3 Singular value spectrum

The first singular value $\sigma_1 = 996$ is approximately twice the second $\sigma_2 = 519$, confirming fast decay and strong low-rank structure in the rating matrix — a favorable property for collaborative filtering.

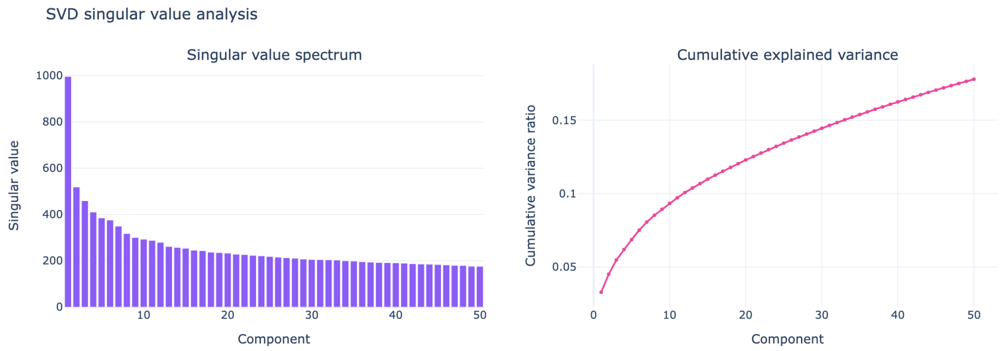


Figure 5: Left: singular value spectrum (first 50 components). Right: cumulative explained variance. Fast decay confirms the matrix has substantial low-rank structure.

### 6.4 SVD score distribution

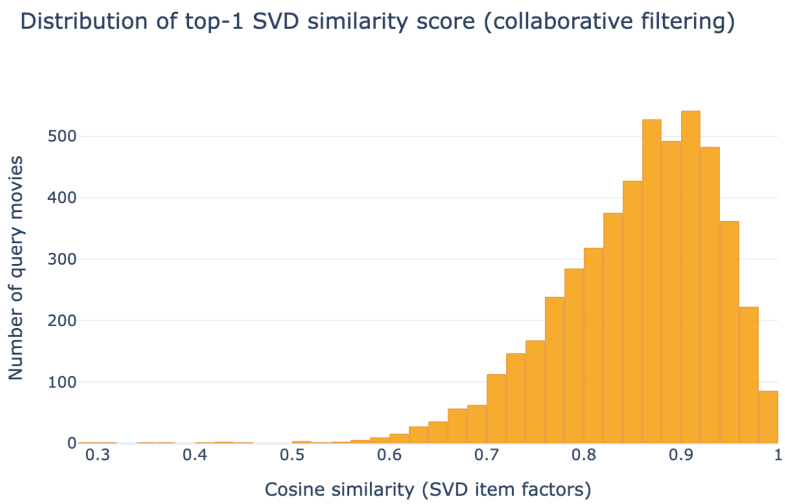


Figure 6: Distribution of top-1 SVD similarity scores across 5,000 query movies. Scores are concentrated at higher values than content-based similarity, reflecting tight co-occurrence structure among popular films.

## 7 Offline Evaluation Protocol

### 7.1 Setup

- **Query movies**: 4,994–4,999 most-rated movies (valid genre labels required)
- **Candidate pool**: top-20 recommendations per system per query
- **Relevance criterion**: shared at least one genre label with the query movie

### 7.2 Relevance criterion

$$\text{relevant}(i, j) = \begin{cases} 1 & \text{if } G_i \cap G_j \neq \emptyset \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

Genre overlap is a reproducible, catalog-level proxy requiring no user ground truth. It favors content methods by construction; a user-history holdout is stronger (see Section 9 for the Leave-One-Out evaluation added after review).

**Recall@$K$ limitation.** In the genre-based protocol, the `Recall@K` denominator was the number of relevant items within the system's own candidate pool — not the full catalog. This inflated Recall artificially (e.g. `content_cosine` Recall@20 = 0.9994). The defect is documented in the `recall_at_k()` function; Section 9 uses **Hit Rate@K** as the correct recall proxy.

### 7.3 Metrics

Table 6: Evaluation metrics

| Metric | Definition |
|---|---|
| Precision@$K$ | $\frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[\text{rec}_k \text{ relevant}]$ |
| Recall@$K$ | $\frac{\lvert\{k \leq K : \text{rec}_k \text{ relevant}\}\rvert}{\lvert\text{all relevant in pool}\rvert}$ |
| NDCG@$K$ | $\frac{\text{DCG@}K}{\text{IDCG@}K}$ where $\text{DCG@}K = \sum_{k=1}^{K} \frac{2^{\text{rel}_k} - 1}{\log_2(k+1)}$ |
| Hit Rate@$K$ | $\mathbf{1}[\exists\, k \leq K : \text{rec}_k \text{ relevant}]$ |

## 8 Evaluation Results

### 8.1 Full metric table

Table 7 presents all metrics at $K \in \{5, 10, 20\}$. Best values per column and $K$ are **bolded**.

Table 7: Offline evaluation results — genre-relevance protocol, $n_{\text{queries}} = 4{,}994$

| System | K | Precision@K | Recall@K | NDCG@K | Hit Rate@K |
|---|---|---|---|---|---|
| popularity_global | 5 | 0.5820 | 0.3621 | 0.8437 | 0.9553 |
| | 10 | 0.5554 | 0.6089 | 0.8390 | 0.9720 |
| | 20 | 0.4593 | 0.9808 | 0.8206 | 0.9808 |
| content_cosine | 5 | **0.9822** | 0.2535 | **0.9930** | **0.9982** |
| | 10 | **0.9775** | 0.5032 | **0.9925** | **0.9990** |
| | 20 | **0.9722** | **0.9994** | **0.9916** | **0.9994** |
| svd_collaborative | 5 | 0.8807 | **0.2656** | 0.9525 | 0.9890 |
| | 10 | 0.8603 | **0.5155** | 0.9496 | 0.9958 |
| | 20 | 0.8379 | 0.9986 | 0.9462 | 0.9986 |

**Key finding.** `content_cosine` leads on all metrics. Its Precision@5 of 0.982 means that 49 out of every 50 recommendations at ranks 1–5 share at least one genre with the query. This reflects the quality of the Week 7 autoencoder, which learned a compact 13-dimensional space highly coherent with MovieLens genre labels.

`svd_collaborative` beats `popularity_global` on Precision@$K$ and NDCG@$K$ but trails `content_cosine`, confirming that behavioral signals add value but content features dominate under genre-based evaluation.

## 8.2 Multi-metric comparison chart

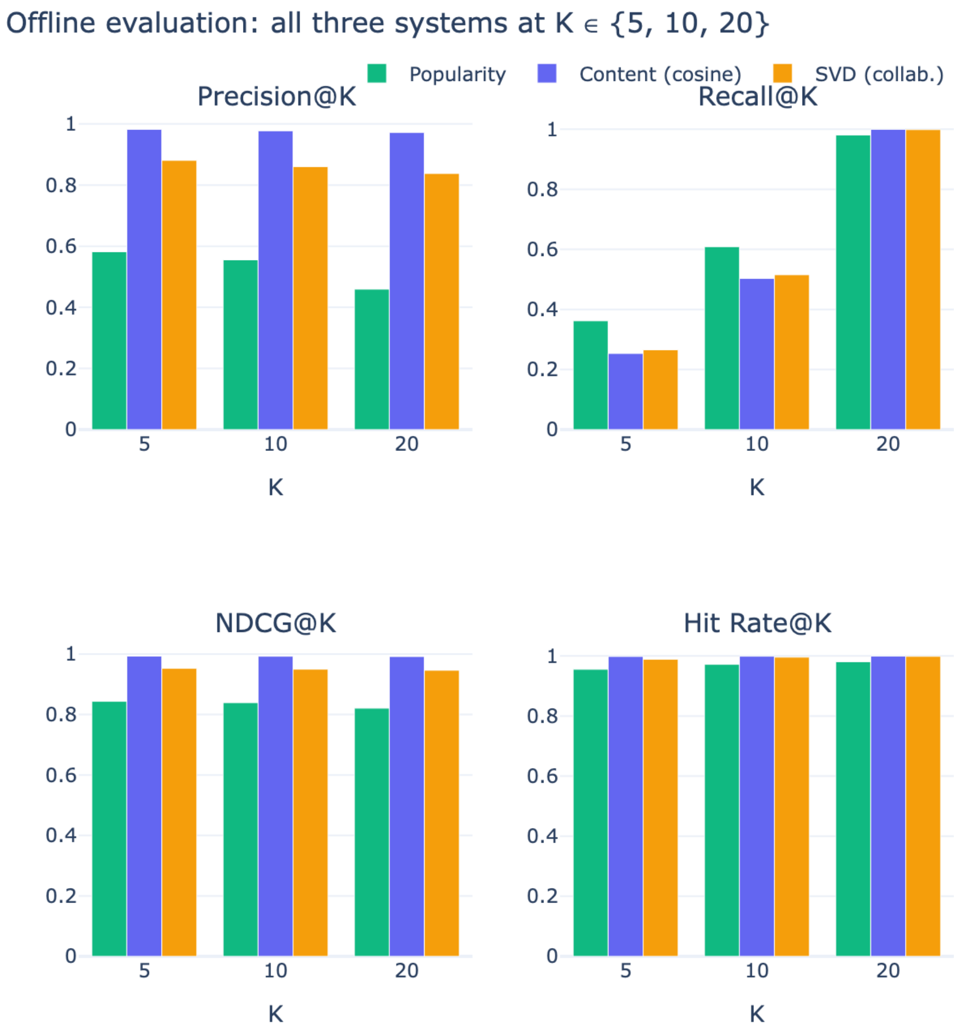


Figure 7: All four metrics for all three systems at $K \in \{5, 10, 20\}$. Content cosine (violet) dominates on Precision and NDCG. SVD collaborative (amber) beats popularity (green) on all metrics.

## 8.3 NDCG@10 summary

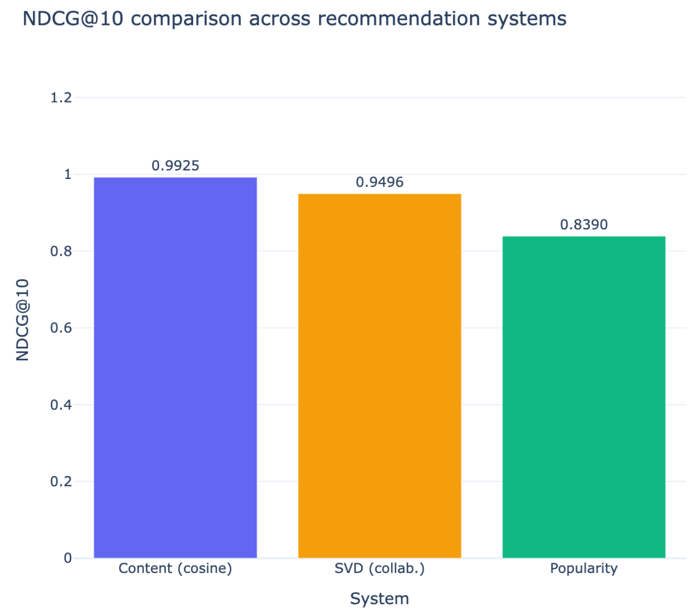


Figure 8: NDCG@10 for the three systems. Content cosine achieves 0.9925, SVD collaborative 0.9496, and popularity global 0.8390.

## 8.4 Per-query NDCG@10 distribution

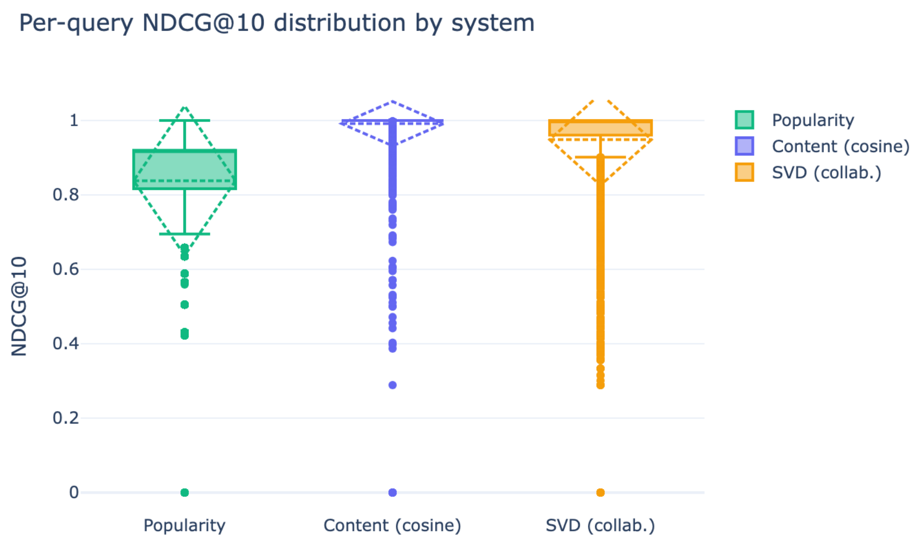


Figure 9: Box plots of per-query NDCG@10 for each system. Content cosine has the highest median and tightest spread. Popularity global shows the widest variance: its performance depends entirely on whether the query genre overlaps with the global top-100.

# 9 Leave-One-Out (LOO) Evaluation

To complement the genre-based protocol, a second evaluation was conducted using user rating histories from `ratings_clean.parquet`.

## 9.1 Protocol

- Users with $\geq 5$ ratings; 10,000 sampled (`seed=42`).
- Last item by timestamp hidden as test; remainder used as training history.
- Each system recommends top-$K$ items excluding already-seen movies.
- Primary metrics: **Hit Rate@**$K$ (equivalent to Recall@$K$ with one test item per user) and **NDCG@**$K$.

*Note on Precision@K in LOO:* with a single test item per user, Precision@$K$ = Hit Rate@$K/K$ by definition and carries no additional information.

## 9.2 Results

Table 8: Leave-One-Out evaluation results ($n$ = 10,000 users, `seed=42`)

| **System** | **Hit Rate@5** | **Hit Rate@10** | **Hit Rate@20** | **NDCG@10** |
|---|---|---|---|---|
| `popularity_global` | 0.0287 | 0.0465 | 0.0767 | 0.0238 |
| `content_cosine` | 0.0219 | 0.0368 | 0.0563 | 0.0187 |
| `svd_collaborative` | **0.0515** | **0.0795** | **0.1221** | **0.0432** |

**Key finding.** `svd_collaborative` **reverses the ranking** seen in the genre-based evaluation: it achieves nearly double the Hit Rate@10 of Popularity (7.95% vs. 4.65%) and more than double that of Content cosine (3.68%). This confirms that SVD captures genuine behavioral predictive signal, while content-based similarity is stronger for catalog discovery but weaker at predicting individual user consumption.

The contrast also validates the Recall correction: the genre-based Recall@20 for `content_cosine` was 0.9994 (inflated by the pool-denominator defect), while the real retrieval rate under LOO is 5.6%.

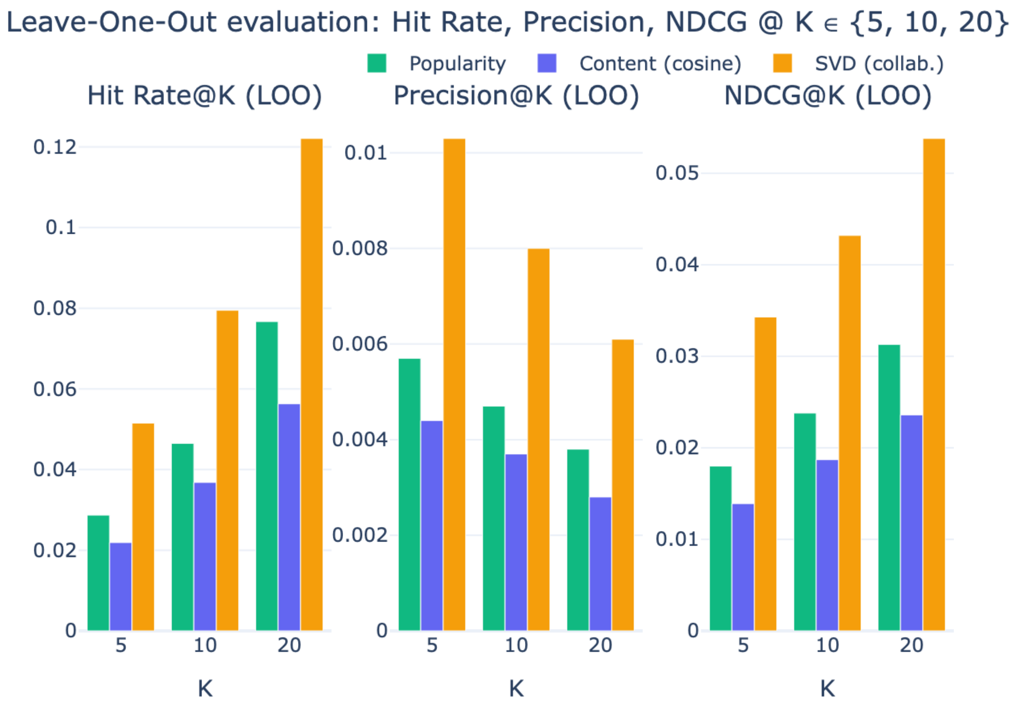


Figure 10: LOO evaluation: Hit Rate, Precision, and NDCG at $K \in \{5, 10, 20\}$. SVD collaborative leads across all metrics; the reversed ranking relative to the genre protocol reflects its stronger user-behavior predictive power.

# 10 Error Analysis

## 10.1 Strong cases (Precision@10 = 1.0)

Table 9: Representative strong-case examples from the error analysis

| System | Query movie | Genres | Reason |
|---|---|---|---|
| popularity_global | Money Train (1995) | Action\|Comedy\|Crime\|Drama\|Thriller | Multi-genre query overlaps with mainstream top-100 |
| popularity_global | Dead Presidents (1995) | Action\|Crime\|Drama | Same |
| content_cosine | Toy Story (1995) | Adventure\|Animation\|Children\|Comedy\|Fantasy | Tight children/animation cluster in AE space |
| content_cosine | Grumpier Old Men (1995) | Comedy\|Romance | Dense comedy-romance neighborhood |
| svd_collaborative | GoldenEye (1995) | Action\|Adventure\|Thriller | Strong user co-occurrence in Action/Thriller |
| svd_collaborative | Tom and Huck (1995) | Adventure\|Children | Clear behavioral cluster among family films |

## 10.2 Failure cases (Hit Rate@10 = 0)

Table 10: Representative failure cases from the error analysis

| System | Query movie | Genres | Root cause |
|---|---|---|---|
| popularity_global | Heidi Fleiss: Hollywood Madam (1995) | Documentary | Top-100 contains no documentaries |
| popularity_global | Anne Frank Remembered (1995) | Documentary | Same |
| content_cosine | Fair Game (1995) | Action | Embedding conflates with non-Action neighbors |
| content_cosine | Repo Man (1984) | Comedy\|Sci-Fi | Mixed-genre embedding pulls to wrong cluster |
| svd_collaborative | Showgirls (1995) | Drama | Anomalous rating pattern; item factors incoherent |
| svd_collaborative | Wyatt Earp (1994) | Western | Few Western films in filtered matrix; noisy factors |

**Pattern.** Popularity global fails systematically for any genre underrepresented in the global top-100 (Documentary, Western, Musical, Film-Noir). Content and SVD fail on edge cases: genre-ambiguous films or items with few ratings.

# 11 What Worked and What Did Not

**What worked.**

- The AE embedding space from Week 7 transfers cleanly to item-to-item recommendation (Precision@10 = 0.978).
- SVD collaborative adds behavioral signal absent from content features — especially useful for genre-pure niche films.
- The evaluation protocol is fully reproducible without user-level ground truth.

**What did not work.**

- **Popularity global**: fails completely for Documentary, Western, Musical, Film-Noir — genres absent from the global top-100.
- **SVD coverage**: 45,871 movies (77.7% of catalog) are excluded from the filtered matrix ($< 50$ ratings). Their item factors would be too noisy.
- **Genre-based evaluation**: favors content methods by construction; SVD may discover more useful behavioral patterns than the metric captures.

# 12 Ethics and Access Note

- **Data source**: GroupLens MovieLens 25M public release (research license).
- **Access**: Educational and research use; raw data not redistributed.
- **Personal data risk**: all user identifiers are anonymized integer IDs.
- **Mitigation**: all analysis at the movie level. SVD item factors are aggregate representations that do not expose individual user behavior.

# 13 Reproducibility

Notebooks must be run in this order:

1. `notebooks/week10/week10_baseline_recommender.ipynb`
2. `notebooks/week10/week10_matrix_factorization.ipynb`
3. `notebooks/week10/week10_offline_evaluation.ipynb`

Table 11: Key parameters for reproducibility

| **Parameter** | **Value** |
|---|---|
| Catalog size | 59,047 movies |
| Global mean rating $\mu$ | 3.0714 |
| Bayesian prior $C$ | 1.0 |
| Embedding dimension | 13 |
| Clusters ($k$) | 7 |
| SVD min. movie ratings | 50 |
| SVD min. user ratings | 20 |
| SVD $n$_components | 50 |
| SVD explained variance | 17.79% |
| Matrix dimensions | 162,540 users $\times$ 13,176 movies |
| Matrix density | 1.15% |
| Random state | 42 |

# 14 Conclusion

Week 10 is complete. Three recommendation systems were built, evaluated, and compared under two complementary offline protocols:

**Genre-relevance evaluation (item-to-item, 4,994 queries).**

- **Content cosine** (NDCG@10 = **0.9925**) is the strongest system for catalog discovery, leveraging the dense autoencoder embedding space from Week 7.
- **SVD collaborative** (NDCG@10 = 0.9496) ranks second, capturing behavioral co-occurrence patterns absent from content features.
- **Popularity global** (NDCG@10 = 0.8390) is the weakest baseline: genre-blind and systematically fails for niche genres.

**Leave-One-Out evaluation (user history, 10,000 users).**

- **SVD collaborative** (Hit Rate@10 = **7.9%**) wins clearly, nearly doubling Popularity (4.6%) and more than doubling Content cosine (3.7%).
- This reversal reveals that SVD is the stronger *predictive* model for user behavior, while content similarity excels at catalog discovery.
- A Recall@$K$ defect in the genre protocol (pool-denominator inflation) was identified and documented; Hit Rate@$K$ under LOO is the corrected metric.

The recommendation layer is now a functioning component of the MovieLens Discovery pipeline. The cluster segmentation from Week 7 underpins the cluster-aware baseline and prepares the ground for the graph analytics layer in Week 12.